



General Engineering

General Engineering Final Year Project Aims and Objectives

Surname: McCleary Forenames: Patrick John

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Student Declaration

I confirm that the work I have submitted is my own original work and that I have made appropriate references to any sources used.

I have read and understood the University guidance on "Unfair Means" (<https://www.sheffield.ac.uk/ssid/exams/plagiarism>), and I confirm that this work complies with those guidelines. I agree to any submitted materials being used by the Department and/or University to detect and take action on any unfair means.

I understand that failure to meet these guidelines may result in one or more of the following penalties being imposed.

1. A reduced mark (including a mark of zero) being recorded for this work.
2. A reduced mark (including a mark of zero) being recorded for the unit/module.
3. A formal reprimand (warning) being placed on my student record at Departmental and/or University level which may be taken into account whenever references are written.
4. Reporting of the incident to the University Disciplinary Committee.
5. Suspension or expulsion from the University.

Student Signature: PMcCleary Date: 15/10/2025

Supervisor Declaration

I have met with the above student to discuss the project and approved the aims and objectives that have been outlined. The assignment will not be accepted without the signature of the Project Supervisor.

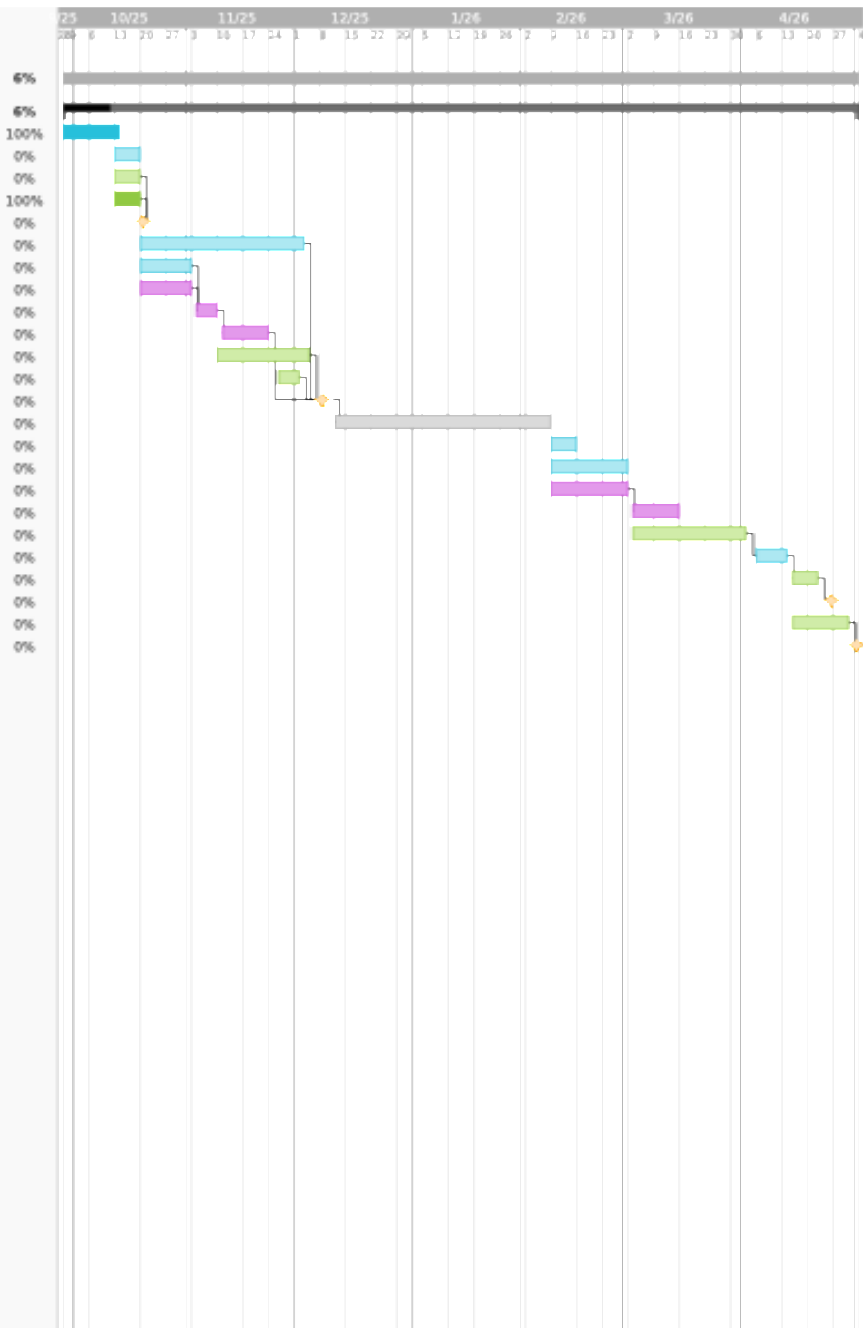
Supervisor signature: Joab Winkler Date: 20/10/2025

Project title
The application of regularisation to common datasets in machine learning
Key aims of the project
• Define the mathematical conditions for which regularisation is necessary using methods of linear algebra such as singular value decomposition and pseudoinverse. • Set up a software framework and implement these criteria using Python. • Test these mathematical criteria on multiple common datasets to categorise the need for regularisation, determining whether it is or is not required. This is essential because its application when not required leads to significant error in the solution. • Verify that the regularised solution satisfies conditions on error and stability.
Tasks to be completed in semester 1
• Conduct literature review on previous related work and research done. • Identify relevant datasets from those that are publicly available. • Set up software environment with Python as the chosen language, Visual Studio Code as the chosen IDE and GitHub for version control and to demonstrate work carried out. • Define, design and plan the experiments to be carried out at first. • Run the initial experiments according to the outlined plan using the chosen datasets and the setup software environment. • Write the interim report throughout the semester and leave time for proof-reading and contingency. • Update the Gantt chart once the scope of the project is more defined to reflect any changes. • Submit interim report.
Tasks to be completed in semester 2
• Meet with supervisor to discuss interim report feedback and decide on an updated plan. • Extend the literature review beyond what was done in first semester the direction of which is dependent on the feedback received. • Run the full experiments whilst gathering crucial data and noting the findings made. • Draft the final report which will expand on the interim report. • Receive feedback on draft final report from supervisor. • Amend final report based on feedback and allow time for proof-reading, formatting and contingency. • Submit final report. • Prepare for oral presentation by making slides which use the created visual aids, rehearsing presentation and considering what questions may be asked. • Deliver oral presentation.

Regularisation

Tasks

- Recap linear algebra
- Learn SVD and pseudoinverse
- Complete aims and objectives
- Make Gantt Chart
- Submit Aims and Objectives
- Literature review
- Dataset and problem selection
- Learn ML tools in Python
- Experimental design
- Initial experiments
- Write interim report
- Update Gantt Chart
- Submit Interim Report
- Revision, Christmas break and exams
- Review interim report feedback
- Extend literature review
- Run full experiments
- Visualise results
- Draft Final Report
- Final Report Feedback
- Proof read and check formatting
- Submit Final Report
- Prepare presentation slides
- Oral Presentation



Legend

- Research
- Report writing
- Technical work and experiments
- Downtime